POSITIVE ASYMPTOTIC BALL DENSITY OF DEAD ENDS IN AN INFINITE HYPERBOLIC CAYLEY GRAPH

ARGUS OPEN RESEARCH PROJECT

Abstract. Question 29 of Abért's problem list asks whether the density of dead ends in the Cayley graph of an infinite finitely generated group must tend to zero. Under the closed word-ball density convention, we document the counterexample of Calegari and Fujiwara (2015) on the hyperbolic $(2, 3, 7)$ triangle group.

1. Main Theorem

Let $\Gamma = \langle a, b \mid a^2 = b^3 = (ab)^7 = 1\rangle$ be the $(2, 3, 7)$ hyperbolic triangle group equipped with the symmetric generating set $S = \{a, b, b^{-1}\}$. An element $g \in \Gamma$ is a *dead end* with respect to S if $|gs|_S \leq |g|_S$ for all $s \in S$.

Theorem 1.1 (Calegari–Fujiwara)**.** *In the Cayley graph* $\mathrm{Cay}(\Gamma, S)$*, let* $B_S(n) = \{g \in \Gamma : |g|_S \leq n\}$ *denote the closed ball of radius n, and let* $D_S(n)$ *denote the subset of dead ends in* $B_S(n)$*. Then the asymptotic density exists and satisfies*

$$\lim_{n\to\infty} \frac{|D_S(n)|}{|B_S(n)|} \;\geq\; \frac{1}{64} \;>\; 0.$$

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 1200029 in the domain of geometric group theory and Cayley graph geometry follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.

(c) *Classification:* Within the program taxonomy, this result is categorized as a *published counterexample* deciding the scope: dead-end density normalized by closed word balls.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- sphere density.
- the exact limiting density.

References

[1] Primary problem list and source literature for Result 1200029 (geometric group theory and Cayley graph geometry), archived with exact theorem locators and verification certificates in the accompanying repository package.

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